INCH-POUND

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SUPERSEDING
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November 12, 1975

FEDERAL SPECIFICATION

ADHESIVE, SYNTHETIC-RUBBER, THERMOPLASTIC, GENERAL-PURPOSE

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers organic solvent base synthetic rubber thermoplastic adhesives for general-purpose use (see 6.1).

1.2 Classification.

1.2.1 Classes. The adhesive covered by this specification shall be of the following classes as specified (see 6.2).

1.2.1.1 Class I. Class 1 - 20 percent nonvolatile content.

1.2.1.1 Class II. Class 2 - 30 percent nonvolatile content (see 6.1).

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2. Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59107 - Toluene, Technical

ENVIRONMENTAL PROTECTION AGENCY

Method 311 - Analysis Of Hazardous Air Pollutant Compounds in Paints and Coatings by Direct Injection into a Gas Chromatograph

(Copies of this document are available online at https://www.epa.gov).

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 100-199 - Rules and Regulations for the Transportation of Hazardous Materials

(Copies of this document are available online at https://www.ecfr.gov/current/title-49).

FEDERAL SPECIFICATIONS

CCC-C-430 - Cloth, Sheeting, Cotton, and Polyester and Cotton

PPP-C-96 - Cans, Metal, 28 Gage and Lighter

PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons)

FEDERAL STANDARDS

FED-STD-123 - Marking for Shipment (Civil Agencies)

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FED-STD-313 - Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities.

(Copies of this document are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of these documents are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM A794/A794M - Standard Specification for Commercial Steel (CS), Sheet, Carbon (0.16 % Maximum to 0.25 % Maximum), Cold-Rolled

ASTM B16 - Standard Specification for Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines

ASTM B209/B209M - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM D86 - Standard Test Method for Distillation of Petroleum Products and Liquid Fuels at Atmospheric Pressure

ASTM D903 - Standard Test Method for Peel or Stripping Strength of Adhesive Bonds

ASTM D1002 - Standard Test Method for Apparent Shear Strength of Single-Lap-Joint Adhesively Bonded Metal Specimens by Tension Loading (Metal-to-Metal)

ASTM D1084 - Standard Test Methods for Viscosity of Adhesives Loading (Metal-to-Metal)

ASTM D1193 - Standard Specification for Reagent Water

ASTM D3272 - Standard Practice for Vacuum Distillation of Solvents from Solvent-Reducible Paints for Analysis

(Copies of these documents are available online at https://www.astm.org/).

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. The adhesive shall be a thermoplastic synthetic rubber base material dissolved in an organic solvent or blend of solvents, modified to the extent that the requirements of this specification are met (see 4.1.1).

3.1.1 Solvents. The manufacturer is given latitude in the selection of solvents or blend of solvents, provided the material meets the requirements specified in 3.1.1.1 and 3.1.1.2.

3.1.1.1 Solvent limitations. The solvent system when tested as specified in 4.3.2.6 shall contain no halogenated compounds nor benzene (benzol) (see 4.3.2.6.2.1 and 4.3.2.6.2.2).

3.1.1.2 Volatiles limitations. The volatile solvent when tested as specified in 4.3.2.6 shall conform by volume to the requirements controlling the emission of solvents into the atmosphere as called out in (a), (b), (c), (d), and (e).

a. A combination of aldehydes or branched chain ketones: 20 percent maximum.
b. A combination of aromatic compounds with eight or more carbon atoms to molecule except ethylbenzene: 8 percent maximum.
c. A combination of ethylbenzene or toluene 20 percent maximum.
d. A combination of solvents with an olefinic or cyclo-olefinic type of unsaturation: 5 percent maximum.
e. Total of (a) + (b) + (c) + (d) = 20 percent maximum.

3.2 Condition in container. When tested as specified in 4.2.5, the adhesive shall be homogeneous and intimately compounded and have no other undesirable qualities.

3.3 Specific gravity. The specific gravity of the adhesive shall be not less than 0.80 nor more than 0.98 when tested as specified in 4.3.2.1.

3.4 Viscosity.

3.4.1 Class 1 viscosity. The viscosity of class 1 adhesive shall be not less than 10 seconds nor more than 40 seconds when tested as specified in 4.3.2.2.

3.4.2 Class 2 viscosity. The viscosity of class 2 adhesive shall be not less than 15 seconds nor more than 45 seconds when tested as specified in 4.3.2.2.

3.5 Nonvolatile content. The nonvolatile content of the adhesive when tested as specified in 4.3.2.3 shall be as follows in table I:

TABLE I. Nonvolatile content limits by class.

Class	Minimum Percent	Maximum Percent
Class 1 (20 percent volatile content)	18	22
Class 2 (30 percent volatile content)	28	32

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3.6 Ash content of nonvolatile matter. The ash content of the nonvolatile matter in the adhesive shall not exceed 18 percent when tested in accordance with 4.3.2.4.

3.7 pH of adhesive-water mixture. The pH of a water adhesive mixture shall be not less than 5.5 nor more than 9.0 when tested in accordance with 4.3.2.5.

3.8 Shear strength. Both classes of adhesive shall conform to the requirements listed in table II when tested in accordance with 4.3.2.7.

TABLE II. Shear strength.

Conditioning time and temperature.	Test temperature	Minimum average value, lbs. per sq. inch	Minimum individual value, lbs. per sq. inch
24 hrs. at 25 ± 2°C (77 ± 3.6°F)	25 ± 2°C (77 ± 3.6°F)	1200	1000
24 hrs. at 77 ± 2°C (170.6 ± 3.6°F), then 2 hrs. at 25 ± 2°C (77 ± 3.6°F)	25 ± 2°C (77 ± 3.6°F)	1200	1000
24 hrs. immersion in toluene at 25 ± 2°C (77 ± 3.6°F)	25 ± 2°C (77 ± 3.6°F)	1000	800
2 hrs. at 50 ± 2°C (122 ± 3.6°F)	50 ± 2°C (122 ± 3.6°F)	700	500

3.9 Peel strength. The adhesive shall have a minimum peel strength of 12 pounds per inch width and minimum individual value of 10 pounds per inch width when tested in accordance with 4.3.2.8.

3.10 Corrosivity. The adhesive shall not be corrosive to steel, aluminum, or brass when tested in accordance with 4.3.2.9.

3.11 Shelf storage life. When the adhesive is stored as specified in 4.3.2.10 there shall be no gelling, curdling, stringing or other nonuniformity. The viscosity shall not have changed more than 10 percent plus or minus from the originally specified viscosity (see 3.4), and the strength values shall be not less than the minimums specified in table II (first condition only) and 3.9 (see 4.1.1).

3.12 Marking.

3.12.1 Civil agencies. In addition to any special marking required by the contract or purchase order, unit and intermediate packages and exterior containers shall be marked in accordance with Fed. Std. No. 123 and with applicable requirements for cement, rubber, of the rules and regulations for the transportation of hazardous materials of the Department of Transportation (see 6.4). Each shipping container shall also be marked by months and year, not code, with the date of manufacture of the adhesive.

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3.12.2 Military agencies. In addition to any special marking required by the contract or purchase order, unit and intermediate packages and exterior containers shall be marked in accordance with MIL-STD-129, and the applicable requirements for cement, rubber, of the rules and requirements for the transportation of hazardous materials of the Department of Transportation (see 6.4). Each shipping container shall also be marked by month and year, not code, with the date of manufacture of the adhesive.

3.12.3 User instruction markings. All primary containers shall be legibly marked or labeled as applicable with the following information:

a. Manufacturer's instructions for use (to include thinning directions if applicable).

b. Date of manufacture (by month and year, not by code)

c. Date of first reinspection (date one year from date of manufacture).

d. Flash point in degrees Centigrade and degrees Fahrenheit.

e. Any special directions for storage or use of the adhesive.

f. In addition to the above the following instructions shall appear on each container:

DANGER! FLAMMABLE
KEEP AWAY FROM HEAT, SPARKS AND OPEN FLAME
STORE AT 60 - 80°F (16 - 27°C)

PRECAUTION: The Surgeon General requires airline respirators to be used unless air sampling shows exposure to be below standards. Then, either chemical cartridge respirators or airline respirators are required. Avoid contact with skin and eyes. Use with adequate ventilation. For other safety recommendations refer to the Safety Data Sheet (SDS). Keep containers closed.

3.13 Toxicity clearance. All new chemicals and materials being added to the Army supply system shall have a toxicity clearance. A toxicity clearance involves a toxicological evaluation of materials prior to introduction into the Army supply system. The Army program manager shall be responsible for identifying technically feasible materials and requesting a toxicity clearance for use of that material within their program (see 6.7).

3.14 SDS. The manufacturer shall comply with the requirements as specified in the OSHA Brief:

(http://www.osha.gov/dsg/hazcom/osha-brief.html).

and, as specified in Appendix D of 29 CFR 1910.1200 see:

(https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1200AppD).

A SDS shall be prepared for the material in accordance with FED-STD-313 and forwarded to the qualifying activity (see 6.4). The SDS shall be included with each shipment of the material covered by this specification and submitted to pertinent Government agencies as stated in FED-STD-313.

3.15 Energetics compatibility (reactivity and storage characteristics). When compatibility for use with a particular explosive or propellant is required, special tests shall be conducted at a

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Government laboratory. The procuring agency shall request that these tests be conducted and shall specify the particular explosive or propellant to be used in the tests (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certification. The supplier shall submit to the contracting officer a certificate of compliance indicating that the adhesive complies with the shelf storage life requirement as specified in 3.11. When certificates of compliance are submitted, the Government reserves the right to check test items to verify the validity of the certificate.

4.2 Quality conformance inspection.

4.2.1 Lot formation. A lot shall consist of adhesive from one batch. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

4.2.2 Sampling. Sampling of the adhesive shall be performed as directed by the procuring agency either (a) in a thoroughly mixed batch of adhesive before packaging, or (b) on a lot of packaged adhesives (filled containers). Sampling for preparation for delivery shall be made on the packaged, or packaged and packed lot.

4.2.3 Examination of filled containers. When the examination is to be performed on filled containers, a random sample of such containers shall be selected for examination from each lot in accordance with MIL-STD-1916 and ANSI/ASQ Z1.4. From these containers 8 ounce samples shall be selected, one from each of two containers, the contents of which appear to have different consistencies. Not less than two samples shall be taken.

4.2.4 Examination of adhesive before packaging. For tests when the sampling is to be performed on the adhesive before packaging, an 8 ounce representative sample shall be selected from each batch.

4.2.5 Condition in container. The condition in the container shall be determined as follows: Each container in the sample shall be opened and sufficient adhesive shall be poured into a clean, dry vessel to permit the remainder of the adhesive to be stirred without spilling. The adhesive shall be stirred with a clean paddle reaching to the bottom of the container. The presence of segregation, caking, lumps, skins or solid particles shall be observed, and differences in consistency between sample units shall be determined by feeling the resistance to movement of the paddle and by observing the material as it drains off the paddle. The material removed shall then be returned to the container.

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4.2.6 Disposition of sample. Sample units selected as specified in 4.2.3 and 4.2.4 shall be tested separately for specific gravity (see 3.3) and viscosity (see 3.4) and all portions must pass both tests. If the sample units pass both tests, the remainder of the samples shall be mixed and this composite Sample used for the other tests. A portion of this composite sample shall be placed in an airtight container and labeled to show the batch number, the name of the material, manufacturer, plant, contract or purchase order number, date of sampling, and identification mark or samples.

4.2.7 Examination. Sample units selected in accordance with 4.2.3 shall be examined for the defects at the acceptable quality level (AQL) shown in table III using ANSI/ASQ Z1.4. This examination is to determine that the adhesive meets the acceptable quality requirements and the preparation for delivery requirements specified in section 5.

TABLE III. Classification of defects.

Material	AQL Percent	Defect Designation Number	Defect	Method of inspection
Adhesive (see 3.2)	2.5	101	Not intimately compounded.	Visual
		102	Not homogeneous.	Visual
Unit container (see 5.1 and 6.8)	2.5	103	Improper fill.	Visual & Scale 2/
		104	Wrong type or size.	Visual
		105	Leakage.	Visual
		106	Improper closure.	Visual
		107	Marking improper or unidentifiable.	Visual
Intermediate package (if required) (see 5.1)	2.5	108	Wrong number of unit containers.	Visual
		109	Wrong type.	Visual
		110	Wrong size.	Visual
		111	Marking improper or unidentifiable.	Visual
Box open (see 5.1)	2.5	112	Wrong type or size.	Visual
		113	Lack of, or improper strapping.	Visual
Box open (see 5.1)	2.5	114	Gross weight, maximum.	Scale 2/
		115	Improperly closed.	Visual
		116	Marking improper or unidentifiable.	Visual

1/ A container filled with the minimum required quantity of adhesive shall be weighed and this weight compared with the weight of unopened containers to determine fill.

2/ Scale shall be approved by the procuring agency.

4.3 Tests.

4.3.1 Classification of tests. All tests shall be performed in accordance with 4.1 and shall be classified as follows:

a. Periodic lot check tests (see 4.3.1.1).
b. Lot acceptance tests (see 4.3.1.2.).

4.3.1.1 Periodic lot check tests. Periodic lot check tests shall be performed on the first lot of adhesive offered to the Government against this specification and once on every 20 lots thereafter, or once every 2 years, whichever is the more frequent, and shall consist of all tests of this

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specification. Whenever a lot is tested and fails to pass one or more of the periodic lot check tests, either regular or recheck, no further lots will be accepted until the supplier has presented sufficient evidence to show that the cause of the failure has been corrected.

4.3.1.2 Lot acceptance tests. Lot acceptance tests shall be made on each lot of product and shall be the basis for acceptance or rejection of the lot. The lot acceptance tests shall consist of the following tests:

a. Specific gravity (see 3.3).
b. Viscosity (see 3.4).

4.3.1.2.1 Certification. Whenever a lot acceptance test is made the supplier shall certify that the lot was made of the same ingredients and by the same process as a previous lot which has passed all tests of the specification. The certificate shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of this agent's authority to bind their authority. The Government reserves the right to check test the adhesive under certification.

4.3.2 Test methods.

4.3.2.1 Specific gravity. The sample shall be thoroughly stirred and the temperature adjusted to 25° ± 2°C (77° ± 3.6°F). The sample shall be poured into a tared 100 milliliter (ml) graduated cylinder to the 100 ml level, with care being taken to eliminate any air bubbles by tapping the cylinder on a large rubber stopper. The cylinder and contents shall be weighed to the nearest 0.1 gram (g). The specific gravity shall be calculated by dividing the weight of the sample by 100 to determine compliance with the applicable requirement of 3.3.

4.3.2.2 Viscosity. The viscosity of the adhesive shall be determined in accordance with ASTM D1084, method A, with consistency cup orifice sizes listed as follows in table IV:

TABLE IV. Consistency cup orifice size by class.

Class	Consistency Cup Orifice Size
Class 1	0.15 inch orifice
Class 2	0.25 inch orifice

4.3.2.3 Nonvolatile content. A weighing bottle (approximately 2.25 inches inside diameter and 1.25 inches deep) shall be heated for l hour at 100° ± 5°C (212° ± 9°F), then cooled in a desiccator, and weighed. All weighing shall be to the nearest one-hundredth gram (g). Approximately 20 g of the sample shall be placed in the prepared weighing bottle, the bottle immediately covered, and then weighed. The cover shall be removed, and the weighing bottle placed on a steam bath until most of the solvent has evaporated. (Caution: Care should be taken to remove as much of the solvent as is practicable by this method in order to eliminate any fire hazard which might otherwise exist when the material is subjected to the subsequent drying operation.) The weighing bottle and contents shall then be placed in an oven at 100° ± 5°C (212° ± 9°F) heated for 5 hours, and the original cover replaced. The weighing bottle contents and cover shall then be cooled in a desiccator and weighed. The heating and weighing procedure shall be continued for heating periods of 30 minutes at 100° ± 5°C (212° ± 9°F) until the loss in weight is less than 0.2 g. The percentage of

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nonvolatile matter shall be calculated by the following formula to determine compliance with the requirements in 3.5

$$\text{Percent of nonvolatile matter} = \frac{A \times 100}{B}$$

where,

A = weight of sample after heating.

B = weight of sample for determination.

4.3.2.4 Ash content of nonvolatile matter. Approximately 20 g of the sample shall be placed in a tared, 150-ml capacity, high-form porcelain crucible. The crucible and contents shall be placed on a steam bath until the solvent has been evaporated. (Caution: Care should be taken to remove as much of the solvent as is practicable by this method in order to eliminate any fire hazard which might otherwise exist when the material is subjected to the subsequent drying operation.) Then the crucible and contents shall be placed in an oven at 100° to 105°C (212° to 221°F) for 5 hours, cooled in a dessicator, and weighed. The weight of the nonvolatile matter in this sample shall be calculated by subtracting the tared weight of the crucible from the combined weight of the nonvolatile matter and the crucible. A 25 ml portion of concentrated nitric acid shall be added to the crucible containing the dried material. The crucible shall be covered with a watchglass to prevent loss by spattering, and heated on an open steam bath until the vigorous reaction which first ensues has subsided. The addition of 25 ml of concentrated nitric acid and heating on the steam bath shall be repeated until no solid material remains in the crucible. The watchglass shall be removed, and the crucible and contents continued to be heated on the open steam bath until all excess nitric acid has been removed. The crucible and contents shall then be placed on an electric hotplate, at medium heat, and heated until all volatile matter has been driven off, and the contents of the crucible have been charred. The crucible and contents shall be ignited to constant weight in a muffle furnace at 600° ± 22°C (1112° + 45°F) The ash content of the nonvolatile matter shall be calculated by means of the following equation to determine compliance with the applicable requirement of 3.6:

$$\text{Percent ash} = \frac{A \times 100}{B}$$

where,

A = weight of ash.

B = weight of nonvolatile matter in sample.

4.3.2.5 pH of adhesive-water mixture.

4.3.2.5.1 Preparation of purified water. The purified water shall consist of 600 ml of ASTM D1193 Type I (ultrapure) with resistivity of ≥18 MΩ-cm (min) at 25°C (77°F) or conductivity <0.056 µS/cm at 25°C (77°F).

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4.3.2.5.2 Determination of pH. Approximately 10 g of the sample shall be placed in a 100 ml. beaker. Then 50 ml of the purified water prepared as specified in 4.3.2.5.1 shall be added, the beaker covered with a watchglass, and the mixture allowed to stand at 73.4° ± 3.6°F (23° + 2°C) for one hour. The mixture shall be stirred every 10 minutes. The electrode of a pH meter (see 6.5), shall be inserted into the beaker so that the mixture of the water and sample covers the active surfaces of the electrodes. The contents of the beaker shall be allowed to remain in contact with the electrodes of the pH meter for 15 minutes before the first measurement is taken. A measurement shall be taken every succeeding 5 minutes until a total of 4 values have been obtained. The pH of the adhesive and water mixture shall be recorded as the average of the 4 Values to determine compliance with the applicable requirements in 3.7.

4.3.2.6 Solvent analysis.

4.3.2.6.1 Separation of volatile portions. Separate volatile portions in accordance with ASTM D3272.

4.3.2.6.2 Solvent composition. The solvent composition shall be determined in accordance with Environmental Protection Agency Air Emission Measurement Center (EMC) method 311 and shall comply with the requirements of 3.1.1.

4.3.2.6.2.1 Halogenated compounds. The presence of halogenated compounds shall be determined using the Beilstein test. The Beilstein test requires copper wire, a Bunsen burner, a sample of the adhesive product, and adequate ventilation to perform. To verify the presence of halogenated compounds, first clean the copper wire by heating it in the Bunsen burner flame to a red incandescence and until the flame no longer shows any green or other colors. Next, touch a sample of the adhesive to the wire and then insert the adhesive dipped copper wire back into the burner flame and observe the flame color emitted. The appearance of a persistent blue or green flame indicates the presence of a halogen (chlorine, bromine, or iodine). To avoid inhalation of fumes, this test procedure should be conducted under a properly operating laboratory hood.

4.3.2.6.2.2 Benzene. When the solvent is tested in accordance with 4.3.2.6.2.1, a trace benzene peak of not more than 2 percent of the toluene peak will be allowed.

4.3.2.7 Shear strength.

4.3.2.7.1 Preparation of shear test specimens. A total of 40 panels shall be prepared from steel conforming to ASTM A794/A794M, grade 1018 or 1020 to measure l by 4 by 0.064 inch. One face of each panel shall be thoroughly cleaned by buffing with fine emery cloth. The panels shall be degreased by immersing them in a solution prepared as follows:

Sodium metasilicate or sodium orthosilicate	30 grams
Sodium alkyl aryl sulfonate	3 grams
Water to make	1 liter

The panels shall be immersed in this solution for 5 minutes dt 60° to 65°C (l40° to 149°F) and rinsed with hot, distilled water, then dried in an oven at 100° to 105°C (212° to 221°F). Using a stiff bristled brush, a thin coating of the adhesive material shall be applied to one end of each of the panels for a length of about 1.25 inches. The adhesive shall be dried in an oven for 1 hour at

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77° ± 2°C (170.6° ± 3.6°F). Then, the 2 panels shall be placed together so that the adhesive coated faces are together, and the panels overlap over an area of exactly 1 square inch. The panels shall he bonded by applying and maintaining a pressure of 500 pounds per square inch, and a temperature of 202° to 207°C (396° to 405°F) for 6 ± l minutes. A hot press shall be used for the bonding.

4.3.2.7.3 Testing of shear specimens. The shear strength tests shall be performed to determine compliance with 3.8. Five of the specimens prepared as specified in 4.3.2.8.1 shall be kept at 25° ± 2°C (77° ± 3.6°F) and 50 ± 4 percent relative humidity for 24 hours and then tested in accordance with ASTM D 1002. A second set of five specimens shall be placed in an oven at 77° ± 2°C (170° ± 3.6°F) for 24 hours. They shall be allowed to cool to 25° ± 2°C (77° ± 3.6°F) and tested as indicated above. A third set of five specimens shall be immersed in toluene complying with the requirements of TT-T-548 at 25° ± 2°C (77° ± 3.6°F) for 24 hours, then wiped dry with a towel and an immediate determination of the shear strength made as indicated above. A fourth set of five specimens shall be placed .in a test chamber heated to 50° ± 2°C (112° ± 3.0°F) for 2 hours, and the shear strength determined at this temperature. The average of each set of five specimens shall be calculated and recorded as the sample value. Storage conditions and testing temperatures for each of the four sets of specimens to prepare for shear testing are outlined for convenience as follows in table V:

TABLE V. Storage conditions and testing temperature.

Specimen Set	Storage Condition	Test Temperature
1	25° ± 2°C (77° ± 3.6°F) and 50 ± 4% RH - 24 hours	25°C
2	77° ± 2°C (170° ± 3.6°F) for 24 hours	25°C
3	Toluene immersion at 77° ± 2°C (170° ± 3.6°F) - 24 hours	25°C
4	50° ± 2°C (122° ± 3.6°F) - 2 hours	50°C

4.3.2.8 Peel strength.

4.3.2.8.1 Preparation of peel test specimens. Adhesive joint specimens are prepared by bonding flexible (cotton) strips to rigid (steel) panels. A total of 10 panels of cold rolled steel measuring l by 8 by 0.0625 inches and a total of 10 strips of cotton cloth conforming to CCC-C-430 and measuring l by 12 inches shall be prepared. The steel panels shall be wiped with a clean cloth dampened with acetone, and then thoroughly vapor degreased with perchloroethylene. The panels shall be allowed to cool to 25° ± 2°C (77° ± 3.6°F). Adhesive shall be applied to a 5 inch length of one area of each steel panel, and to a 5 inch length of one side of each cotton cloth strip. The number of coats of adhesive and the drying time between shall be as specified by the manufacturer, but no more than 2 coats applied to each faying surface. The adhesive shall not be thinned. All cementing operations shall be made at 25° ± 2°C (77° ± 3.6°F) and 50 ± 4 percent relative humidity. The bonding surfaces shall be brought together immediately after coating with adhesive. The unhanded ends of the adherends shall be adjacent to each other. Hand pressure shall be maintained to keep the faying surfaces in intimate contact, and all wrinkles shall be removed. The specimens shall be air dried for 18 hours at 25° ± 2°C (77° ± 3.6°F), then dried for 24 hours at 50° ± 2°C (122° ± 3.6°F) in an air-circulating oven.

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4.3.2.8.2 Testing of peel specimens. The peel specimens prepared in 4.3.2.8.1 shall be conditioned at 25° ± 2°C (77° ± 3.6°F) and 50 ± 4 percent relative humidity for 4 hours. The specimens shall be tested in accordance with ASTM D903. The average shall be calculated and recorded as the sample value.

4.3.2.9 Corrosivity. Test panels of steel, aluminum, and brass conforming to ASTM A794/A794M, (1018 or 1020), ASTM B209/B209M, and Copper Alloy Number 360 under ASTM B16 shall be buffed with fine emery cloth and rinsed free from residual abrasive, dirt, and or oil using N-Propyl Bromide (1-Bromopropane) solvent 1/. A few drops of the adhesive shall then be applied to each panel and allowed to dwell for 24 hours at normal room temperature. After the dwell period, the compound shall be removed by wiping with a cloth wet with the solvent and the sheets shall then be examined for etching. Anything more than a superficial discoloration of the panel surfaces shall be considered evidence of corrosivity.

1/ The solvent trichloroethylene (TCE), previously required under MMM-A-189C for determination of corrosivity, was banned by the US EPA on 17 December 2024 due to health hazards. The phase out period to use remaining TCE stock concluded on 17 March 2025 and TCE is now prohibited for commercial and industrial use.

4.3.2.10 Shelf storage life verification. A one pint container, or an equivalent amount of smaller sized containers, of the adhesive shall be stored in the original unopened container or containers for one year from the date of manufacture at 75° ± 5°F (24° ± 3°C) and then shall be tested to determine compliance with the requirement of 3.11.

4.3.2.11 Rejection criteria. Failure of any test specimen or sample to meet the test requirements specified herein shall be cause for rejection of the lot represented.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Markings. Containers of adhesive shall include labels that identify the specification number, and class so that traceability of the lot can be verified and include a SDS (see 3.12 and 3.14).

5.3 Packaging and container notes. Due to the substantial passage of time since the last revision of this specification, additional packaging and container information that may be helpful pertaining to sizes, materials and coloration that was listed under the prior revision are included in 6.8 and its sections therein. This legacy content will be deleted in the next revision or amendment to this specification.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. While the adhesive classes in this specification are intended for general purpose applications, the user should verify their suitability by preparing and testing specimens with the intended adherends, taking care that the surfaces of such adherends are properly prepared prior to bonding. Surface preparation methods are detailed in the technical literature and can often be obtained as recommendations by adhesive manufacturers for their particular formulation. The adhesives, being thermoplastic, are not recommended for applications where they will be subject to either continuous high stress, or stress at an elevated temperature. The class 2 adhesive (see 1.2) with its higher solids content forms heavier films with controlled penetration and is used for porous surfaces.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Class required (see 1.2).
c. Size of container required (see 5.1).
d. Maximum shelf life recommended by manufacturer.
e. Selection of the applicable levels of packaging and packing (5.1).
f. Special labelling and marking if required (see 3.14.1, 4.27, 5.2, and 6.6).
g. Whether the adhesive must be compatible with explosives or propellants (see 3.15 and 6.3).
h. Preparation of SDS in accordance with FED-STD-313 for the adhesive and inclusion of SDS with shipment of material (see 3.14 and 6.6).

6.3 Compatibility (reactivity and storage characteristics). For information and tests on the compatibility of adhesives (otherwise meeting the requirements of this specification) with explosives or propellants, contact U.S. Army Armaments Center, Materials Analysis Branch, FCDD-ACM-AP, Picatinny Arsenal, NJ 07806. Testing can be carried out for a fee.

6.4 Department of Transportation regulations. As of the date of this revision, 49 CFR 100-199, “Rules and Regulations for the Transportation of Hazardous Materials”, FED-STD-123, and FED-STD-313 cover the transportation of hazardous materials other than explosives and other dangerous articles.

6.5 pH test equipment. Almost any commercially available pH meter will meet the requirements of 4.3.2.5.2. Various commercial scientific equipment vendors offer a wide variety of suitable models. Most portable instruments are also acceptable (see 4.3.2.5.2).

6.6 SDS. Contracting officers will identify those activities requiring copies of a completed SDS, when implemented, prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

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6.7 Toxicity clearance request. Department of the Army Regulation (AR) 40-5, Preventive Medicine, (AR) 70-1, Acquisition Policy, and Department of the Army Pamphlet 70-3, Acquisition Procedures, require a toxicity clearance. Army toxicity questions and/or a toxicity clearance request should be addressed to: DHA Public Health, ATTN: DCPH-ATS-TOX, 8300 Ricketts Point Road, APG-EA, MD 21010-5403 or emailed to dha.apg.Pub-Health-A.mbx.tox-info@health.mil.

6.8 Packaging and container considerations. Due to the substantial length of time since the last revision of this specification, additional packaging and container information are included as a courtesy in this section.

6.8.1 Container sizes and materials. Adhesive furnished in 0.75-ounce unit quantities should be contained in glass bottles with a screw cap. The 1-pint, 1-quart, or 1-gallon should be contained in metal cans complying with the requirements of PPP-C-96, type v, class 2, or in 5-gallon steel pails complying with the requirements of PPP-P-704, type II, class 3, with olive drab exterior coating, as applicable. Closure of 1-gallon containers shall be in accordance with the appendix to PPP-C-96. If requested in the contract or purchase order, the adhesive may be packaged in the quantities as specified (see 6.2) and in accordance with the manufacturer's commercial practice. However, protection shall be such as to prevent deterioration during shipment and prior to use and packed in a manner to ensure carrier acceptance and safe delivery at destination in accordance with Department of Transportation regulations.

6.9 Subject term (key word) listing.

Adhesion	Construction	Glue
Bonding	Cure	

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Air Force - 84

Preparing activity:
ARMY - MR

(Project 8040-2025-006)

Review activities:
Army - MD, MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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